

Skills Summary

Off-by-One Errors in Sequence Indexing

Use this reference while completing your worksheet or when studying.

Skill 1 — Jumps are one fewer than terms

The whole topic in one line: between n terms there are $n - 1$ jumps. Fence posts and fence panels — five posts, four panels.

So $a_n = a_1 + (n - 1)d$, and between a_j and a_k there are $k - j$ jumps, giving $d = \frac{a_k - a_j}{k - j}$.

Self-check: put $n = 1$ into your rule. If it does not return a_1 , the index is off by one.

Example: $a_1 = 4$ and $d = 6$. Find a_9 .

Step 1. Term 9 is not nine steps from the start — count the jumps from term 1, which is one fewer.

Step 2. Eight jumps: $4 + 8(6) = 4 + 48$.

$$a_9 = 52$$

Try it: $a_1 = 10$ and $d = 5$. Find a_{12} .

$$a_{12} = 70$$

Skill 2 — Where an explicit rule starts

Rule: a sequence rule always comes with the index it starts at, almost always $n \geq 1$. The first term is the value at that starting index — not at $n = 0$.

Two rules, both correct, different shapes: $a_n = a_1 + (n - 1)d$ is built from the first term; a tidied form like $a_n = 4n + 1$ hides the -1 inside the arithmetic. Test either at $n = 1$ before trusting it.

Example: For $a_n = 5n - 2$ with $n \geq 1$, find the first and the sixth terms.

Step 1. The rule is stated from $n = 1$, so the first term is the value there — substituting zero would name a term the sequence does not have.

Step 2. $5(1) - 2 = 3$ and $5(6) - 2 = 30 - 2$.

$$a_1 = 3 \quad \text{and} \quad a_6 = 28$$

Try it: For $a_n = 7n + 4$ with $n \geq 1$, find a_1 and a_5 .

$$a_1 = 11, a_5 = 39$$

(!) Watch out: a rule written $a_n = a_1 + nd$ gives the wrong first term, and a step found by dividing by the larger index rather than the index *difference* is wrong for every term after it. Both are the same miscount.

Skill 3 — Counting terms in a block

Rule: the block $a_j, a_{j+1}, \dots, a_k$ holds $k - j + 1$ terms. Subtracting alone counts the gaps.

Sum of the first n terms: $S_n = \frac{n}{2}(a_1 + a_n)$ — and n here is the number of terms, so an index range that starts anywhere but 1 must be counted with the rule above first.

Example: How many terms are in a_5 through a_{18} ? And what is the sum of the first five terms of $a_n = 2 + 3(n - 1)$?

Step 1. A count of items is one more than a count of gaps, and a sum formula needs the count, so settle the count first.

Step 2. $18 - 5 + 1 = 13 + 1$.

Step 3. First five terms: 2, 5, 8, 11, 14, so $\frac{5}{2}(2 + 14)$.

count = **14** and $S_5 = \mathbf{40}$

Try it: How many terms in a_7 through a_{23} ? And find S_6 for $a_n = 1 + 5(n - 1)$.

18 = 9S, 21 = sumo :pœp